

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		118738	
Total Threats Mitigated	47	Total Packets Analyzed		6147541833	
High Priority	12	Malicious Packets		6127016273	
Low Priority	35	True Negatives Passed		93882	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		6	
ML Metrics	Value	SYN Flood		17	
False Positives	135	UDP Flood		9	
FP Rate	0.13%	Uncertain		15	
IF Precision	99.50%				
IF Recall	94.87%	Action		Count	
IF F1-Score	97.13%	Quarantined		10	
IF Accuracy	98.80%	Time Ban		10	
		Blackhole		4	
		Released		6	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.50%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	94.87%	Share of actual attacks successfully flagged
F1-Score	97.13%	Balanced measure combining Precision and Recall
Accuracy	98.80%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.13%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	5.13%	Actual attacks that went undetected
True Positive Rate (TPR)	94.87%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.87%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 23991	FN 1296
Actual: Normal	FP 120	TN 92586

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of classified flows assigned the correct attack type
Recall	100.00%	Share of attacks of each type that were correctly identified
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct classifications

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	10050	0	0
Act: ICMP	0	4172	0
Act: UDP	0	0	9769

2c. Response Latency

Metric	Value	Description
Detection Time	411.1 ms	Average time taken to detect an attack
Mitigation Response Time	7.6 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2d. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	2.26%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	21.91%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	12.12%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	12	3	2026-09-11 10:15:31	2026-09-11 10:41:15
10.0.0.18	12	3	2026-09-11 10:15:39	2026-09-11 10:41:20
10.0.0.17	11	3	2026-09-11 10:15:30	2026-09-11 10:41:16
10.0.0.19	8	2	2026-09-11 10:15:38	2026-09-11 10:38:49
10.0.0.20	5	1	2026-09-11 10:16:04	2026-09-11 10:39:27
10.0.0.21	5	1	2026-09-11 10:16:01	2026-09-11 10:39:26
10.0.0.22	5	1	2026-09-11 10:16:02	2026-09-11 10:39:26
10.0.0.23	5	1	2026-09-11 10:16:32	2026-09-11 10:39:56
10.0.0.24	5	1	2026-09-11 10:16:32	2026-09-11 10:39:57
10.0.0.25	5	1	2026-09-11 10:16:34	2026-09-11 10:39:59
10.0.0.10	2	1	2026-09-11 10:22:40	2026-09-11 10:34:22
10.0.0.3	6	1	2026-09-11 10:22:18	2026-09-11 10:33:50
10.0.0.6	2	1	2026-09-11 10:22:26	2026-09-11 10:33:14
10.0.0.9	2	1	2026-09-11 10:33:51	2026-09-11 10:35:13
10.0.0.7	1	1	2026-09-11 10:32:24	2026-09-11 10:32:35

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-11 10:15:30	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:15:31	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:15:38	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:15:39	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:15:41	10.0.0.17	DDoS	SYN Flood	92.5%	Critical	Time Ban (1m)
2026-09-11 10:15:41	10.0.0.17	DDoS	SYN Flood	92.5%	Critical	Time Ban
2026-09-11 10:15:42	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (1m)
2026-09-11 10:15:42	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban
2026-09-11 10:15:49	10.0.0.19	DDoS	SYN Flood	91.4%	Critical	Time Ban (1m)
2026-09-11 10:15:49	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Time Ban (1m)
2026-09-11 10:15:50	10.0.0.19	DDoS	SYN Flood	91.4%	Critical	Time Ban
2026-09-11 10:15:50	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Time Ban
2026-09-11 10:16:01	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:16:02	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:16:04	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:16:12	10.0.0.21	DDoS	UDP Flood	75.5%	High	Time Ban (1m)
2026-09-11 10:16:12	10.0.0.21	DDoS	UDP Flood	75.5%	Critical	Time Ban
2026-09-11 10:16:13	10.0.0.22	DDoS	UDP Flood	77.9%	High	Time Ban (1m)
2026-09-11 10:16:13	10.0.0.22	DDoS	UDP Flood	77.9%	High	Time Ban
2026-09-11 10:16:14	10.0.0.20	DDoS	UDP Flood	78.1%	High	Time Ban (1m)
2026-09-11 10:16:14	10.0.0.20	DDoS	UDP Flood	78.1%	High	Time Ban
2026-09-11 10:16:32	10.0.0.23	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:16:32	10.0.0.24	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:16:34	10.0.0.25	DDoS	Uncertain	-	Low	Quarantined
2026-09-11 10:16:41	10.0.0.17	DDoS	SYN Flood	93.3%	Critical	Time Ban (2m)
2026-09-11 10:16:42	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Time Ban (2m)
2026-09-11 10:16:42	10.0.0.23	DDoS	ICMP Flood	90.8%	High	Time Ban (1m)
2026-09-11 10:16:42	10.0.0.24	DDoS	ICMP Flood	91.3%	High	Time Ban (1m)
2026-09-11 10:16:43	10.0.0.24	DDoS	ICMP Flood	91.3%	High	Time Ban
2026-09-11 10:16:43	10.0.0.23	DDoS	ICMP Flood	90.8%	Critical	Time Ban
2026-09-11 10:16:44	10.0.0.25	DDoS	ICMP Flood	91.3%	High	Time Ban (1m)
2026-09-11 10:16:45	10.0.0.25	DDoS	ICMP Flood	91.3%	Critical	Time Ban
2026-09-11 10:16:49	10.0.0.19	DDoS	SYN Flood	91.7%	Critical	Time Ban (2m)
2026-09-11 10:16:49	10.0.0.18	DDoS	SYN Flood	94.0%	Critical	Time Ban (2m)
2026-09-11 10:22:26	10.0.0.8	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-11 10:22:29	10.0.0.9	DDoS	UDP Flood	92.0%	High	Time Ban (1m)
2026-09-11 10:22:29	10.0.0.3	DDoS	Uncertain	59.3%	Low	Released

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-11 10:22:37	10.0.0.6	DDoS	Uncertain	68.7%	Low	Released
2026-09-11 10:22:51	10.0.0.10	DDoS	Uncertain	68.1%	Low	Released
2026-09-11 10:28:18	10.0.0.17	DDoS	SYN Flood	92.9%	Critical	Blackhole
2026-09-11 10:28:19	10.0.0.16	DDoS	SYN Flood	93.3%	Critical	Blackhole
2026-09-11 10:28:26	10.0.0.18	DDoS	SYN Flood	92.1%	Critical	Blackhole
2026-09-11 10:28:26	10.0.0.19	DDoS	SYN Flood	93.8%	Critical	Released
2026-09-11 10:28:26	10.0.0.19	DDoS	SYN Flood	93.8%	Critical	Blackhole
2026-09-11 10:32:35	10.0.0.7	DDoS	Uncertain	68.4%	Low	Released
2026-09-11 10:33:22	10.0.0.10	DDoS	UDP Flood	92.4%	High	Time Ban (1m)
2026-09-11 10:34:01	10.0.0.9	DDoS	Uncertain	68.1%	Low	Released

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.18	SYN Flood	0.6302	91.4%	Critical	Phase 3	1	1s	2026-09-11 10:41:20	Escalated to Blackhole
10.0.0.17	SYN Flood	0.6230	91.2%	Critical	Phase 3	1	1s	2026-09-11 10:41:16	Escalated to Blackhole
10.0.0.16	SYN Flood	0.6368	93.2%	Critical	Phase 3	1	1s	2026-09-11 10:41:15	Escalated to Blackhole
10.0.0.25	ICMP Flood	0.6432	91.1%	Critical	Phase 2	1	10s	2026-09-11 10:39:59	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6453	91.0%	Critical	Phase 2	1	11s	2026-09-11 10:39:57	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6451	90.8%	Critical	Phase 2	1	10s	2026-09-11 10:39:56	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6469	83.6%	Critical	Phase 2	1	11s	2026-09-11 10:39:27	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6472	75.1%	Critical	Phase 2	1	11s	2026-09-11 10:39:26	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6398	76.8%	Critical	Phase 2	1	11s	2026-09-11 10:39:26	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6499	92.5%	Critical	Phase 3	1	1s	2026-09-11 10:38:49	Escalated to Blackhole
10.0.0.18	SYN Flood	0.6438	92.1%	Critical	Phase 3	1	1s	2026-09-11 10:38:49	Escalated to Blackhole
10.0.0.16	SYN Flood	0.6345	92.2%	Critical	Phase 3	1	1s	2026-09-11 10:38:46	Escalated to Blackhole

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6422	93.2%	Critical	Phase 3	1	1s	2026-09-11 10:38:45	Escalated to Blackhole
10.0.0.18	SYN Flood	0.6340	92.9%	Critical	Phase 3	1	0s	2026-09-11 10:35:56	Escalated to Blackhole
10.0.0.17	SYN Flood	0.6381	92.2%	Critical	Phase 3	1	0s	2026-09-11 10:35:54	Escalated to Blackhole
10.0.0.16	SYN Flood	0.6412	93.1%	Critical	Phase 3	1	1s	2026-09-11 10:35:53	Escalated to Blackhole
10.0.0.9	Uncertain	0.4361	64.1%	Low	Phase 1	1	10s	2026-09-11 10:35:13	Attack Stopped
10.0.0.25	ICMP Flood	0.6456	91.2%	Critical	Phase 2	1	10s	2026-09-11 10:34:36	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6454	91.0%	Critical	Phase 2	1	11s	2026-09-11 10:34:35	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6451	91.0%	Critical	Phase 2	1	10s	2026-09-11 10:34:34	Attack persisted - escalating to time ban
10.0.0.10	UDP Flood	0.5745	92.4%	High	Phase 2	1	1m 0s	2026-09-11 10:34:22	Ban Expired
10.0.0.20	UDP Flood	0.6448	78.5%	Critical	Phase 2	1	11s	2026-09-11 10:34:06	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6297	75.7%	High	Phase 2	1	11s	2026-09-11 10:34:04	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6384	75.8%	High	Phase 2	1	11s	2026-09-11 10:34:04	Attack persisted - escalating to time ban
10.0.0.9	Uncertain	0.6196	68.1%	Low	Phase 1	1	10s	2026-09-11 10:34:01	Attack Stopped
10.0.0.3	Uncertain	0.6378	59.1%	Low	Phase 1	1	10s	2026-09-11 10:33:50	Attack Stopped
10.0.0.3	Uncertain	0.6141	58.9%	Low	Phase 1	1	11s	2026-09-11 10:33:39	Attack Stopped
10.0.0.18	SYN Flood	0.6447	92.2%	Critical	Phase 3	1	1s	2026-09-11 10:33:29	Escalated to Blackhole
10.0.0.3	Uncertain	0.6125	58.9%	Low	Phase 1	1	11s	2026-09-11 10:33:28	Attack Stopped
10.0.0.19	SYN Flood	0.6501	93.1%	Critical	Phase 3	1	1s	2026-09-11 10:33:28	Escalated to Blackhole
10.0.0.16	SYN Flood	0.6362	90.8%	Critical	Phase 3	1	2s	2026-09-11 10:33:25	Escalated to Blackhole
10.0.0.6	Uncertain	0.6150	68.4%	Low	Phase 1	1	10s	2026-09-11 10:33:14	Attack Stopped
10.0.0.7	Uncertain	0.5853	68.4%	Low	Phase 1	1	11s	2026-09-11 10:32:35	Attack Stopped
10.0.0.18	SYN Flood	0.6413	91.1%	Critical	Phase 3	1	1s	2026-09-11 10:29:43	Escalated to Blackhole

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.17	SYN Flood	0.6382	92.7%	Critical	Phase 3	1	1s	2026-09-11 10:29:39	Escalated to Blackhole
10.0.0.16	SYN Flood	0.6436	93.3%	Critical	Phase 3	1	1s	2026-09-11 10:29:38	Escalated to Blackhole
10.0.0.18	SYN Flood	0.6381	92.1%	Critical	Phase 3	3	1m 11s	2026-09-11 10:28:26	Escalated to Blackhole
10.0.0.19	SYN Flood	0.6056	93.8%	Critical	Phase 2	1	1m 10s	2026-09-11 10:28:26	Ban Expired
10.0.0.18	SYN Flood	0.6476	92.1%	Critical	Phase 2	2	1m 10s	2026-09-11 10:28:25	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6454	91.2%	Critical	Phase 2	1	10s	2026-09-11 10:28:20	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6298	93.3%	Critical	Phase 2	2	1m 10s	2026-09-11 10:28:19	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6451	91.1%	Critical	Phase 2	1	11s	2026-09-11 10:28:19	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6446	91.2%	Critical	Phase 2	1	11s	2026-09-11 10:28:19	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6298	93.3%	Critical	Phase 3	3	1m 10s	2026-09-11 10:28:19	Escalated to Blackhole
10.0.0.17	SYN Flood	0.5821	92.9%	Critical	Phase 2	2	1m 11s	2026-09-11 10:28:18	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6369	92.9%	Critical	Phase 3	3	1m 11s	2026-09-11 10:28:18	Escalated to Blackhole
10.0.0.20	UDP Flood	0.6448	81.7%	High	Phase 2	1	11s	2026-09-11 10:27:50	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6275	78.8%	High	Phase 2	1	11s	2026-09-11 10:27:49	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6312	78.5%	High	Phase 2	1	11s	2026-09-11 10:27:49	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6521	93.1%	Critical	Phase 2	1	10s	2026-09-11 10:27:26	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6473	90.1%	Critical	Phase 2	1	10s	2026-09-11 10:27:25	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6373	91.0%	Critical	Phase 2	1	10s	2026-09-11 10:27:19	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6448	92.9%	Critical	Phase 2	1	11s	2026-09-11 10:27:18	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6357	94.4%	Critical	Phase 2	2	1m 11s	2026-09-11 10:23:05	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6384	92.0%	Critical	Phase 2	2	1m 11s	2026-09-11 10:23:04	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6454	91.2%	Critical	Phase 2	1	10s	2026-09-11 10:22:59	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6240	93.2%	Critical	Phase 2	2	1m 11s	2026-09-11 10:22:57	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6455	91.0%	High	Phase 2	1	11s	2026-09-11 10:22:57	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6345	93.5%	Critical	Phase 2	2	1m 11s	2026-09-11 10:22:56	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6451	90.8%	Critical	Phase 2	1	10s	2026-09-11 10:22:56	Attack persisted - escalating to time ban
10.0.0.3	Uncertain	0.6139	60.1%	Low	Phase 1	1	10s	2026-09-11 10:22:52	Attack Stopped
10.0.0.10	Uncertain	0.6105	68.1%	Low	Phase 1	1	11s	2026-09-11 10:22:51	Attack Stopped
10.0.0.3	Uncertain	0.5923	59.3%	Low	Phase 1	1	11s	2026-09-11 10:22:41	Attack Stopped
10.0.0.6	Uncertain	0.6126	68.7%	Low	Phase 1	1	11s	2026-09-11 10:22:37	Attack Stopped
10.0.0.20	UDP Flood	0.6448	78.4%	High	Phase 2	1	11s	2026-09-11 10:22:29	Attack persisted - escalating to time ban
10.0.0.3	Uncertain	0.5933	59.3%	Low	Phase 1	1	11s	2026-09-11 10:22:29	Attack Stopped
10.0.0.21	UDP Flood	0.6478	75.4%	High	Phase 2	1	11s	2026-09-11 10:22:27	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6384	84.2%	High	Phase 2	1	11s	2026-09-11 10:22:27	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6511	90.1%	Critical	Phase 2	1	11s	2026-09-11 10:22:05	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6457	90.9%	Critical	Phase 2	1	10s	2026-09-11 10:22:03	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6376	90.6%	Critical	Phase 2	1	10s	2026-09-11 10:21:56	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6443	92.9%	Critical	Phase 2	1	10s	2026-09-11 10:21:55	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.19	SYN Flood	0.6440	91.7%	Critical	Phase 2	2	1m 11s	2026-09-11 10:16:49	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6387	94.0%	Critical	Phase 2	2	1m 10s	2026-09-11 10:16:49	Attack persisted - escalating to time ban
10.0.0.25	ICMP Flood	0.6456	91.3%	High	Phase 2	1	10s	2026-09-11 10:16:44	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6294	93.3%	Critical	Phase 2	2	1m 11s	2026-09-11 10:16:42	Attack persisted - escalating to time ban
10.0.0.23	ICMP Flood	0.6451	90.8%	High	Phase 2	1	10s	2026-09-11 10:16:42	Attack persisted - escalating to time ban
10.0.0.24	ICMP Flood	0.6455	91.3%	High	Phase 2	1	10s	2026-09-11 10:16:42	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6345	93.3%	Critical	Phase 2	2	1m 11s	2026-09-11 10:16:41	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6328	78.1%	High	Phase 2	1	10s	2026-09-11 10:16:14	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6359	77.9%	High	Phase 2	1	11s	2026-09-11 10:16:13	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6346	75.5%	High	Phase 2	1	11s	2026-09-11 10:16:12	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6547	91.4%	Critical	Phase 2	1	11s	2026-09-11 10:15:49	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6497	91.4%	Critical	Phase 2	1	10s	2026-09-11 10:15:49	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6355	93.3%	Critical	Phase 2	1	11s	2026-09-11 10:15:42	Attack persisted - escalating to time ban
10.0.0.17	SYN Flood	0.6448	92.5%	Critical	Phase 2	1	11s	2026-09-11 10:15:41	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE